MARUTHAPPAN M

Software Developer
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LinkedIn | GitHub

PROFESSIONAL SUMMARY

B.Tech final-year Information Technology student with strong knowledge of Data Structures & Algorithms, Java, Spring Boot, and RESTful API development, plus hands-on experience with relational databases, debugging, and testing through internships and projects. Built backend-driven applications including an AI-powered semantic search system and an AI-based fraud detection platform. Seeking a Software Developer role to write clean, scalable code, collaborate in Agile teams, and grow as a full stack engineer.

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, JavaScript
Backend & Frameworks: Spring Boot, Node.js, Express.js, RESTful APIs, API Integration, Backend Development
Databases: SQL, CRUD Operations, Database Fundamentals, PostgreSQL, SQLite
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks
Software Development: SDLC, Agile, Debugging, Software Testing, Troubleshooting, Log Analysis, Android Development
AI & Tools: Gemini API, Groq API, Git, GitHub, Linux, VS Code, Android Studio

PROJECTS

SnapSense | *Java, Spring Boot, SQLite, OCR*

- Developed an AI-powered screenshot search app using OCR, semantic tagging, and multilingual support with Java, Spring Boot, and SQLite.
- Implemented semantic search, duplicate detection, and parallel processing to improve search accuracy and speed.
- Debugged and tested the application workflow end to end to improve reliability.

NexusGuard – AI-Powered Financial Fraud Detection System | *Python, Agentic AI, GNN, Isolation Forest, Gemini API, Groq API*

- Built an AI-driven fraud detection system to identify suspicious transactions, money-laundering patterns, and potential fraud rings.
- Developed three AI agents for investigation, fraud-ring analysis, and decision support using graph-based and anomaly-detection techniques.
- Integrated Gemini and Groq APIs to provide GenAI-assisted fraud investigation and generate explainable analysis summaries.

MeshWave | *ESP32, BLE, LoRa*

- Built an ESP32-based offline messaging prototype using BLE and LoRa for device-to-device communication without internet or cellular networks.
- Debugged and tested firmware, connectivity, and network issues to improve system reliability and performance.
- Led a four-member team at the VIT Makeathon, applying system design, problem-solving, and team collaboration.

INTERNSHIPS

Python Development Intern – Cognifyz Technologies

- Wrote Python code for automation and software development tasks, with hands-on debugging, testing, and issue resolution.
- Strengthened programming logic, debugging, and systematic problem-solving skills.

Generative AI Intern – BigLearn

- Worked on SnapSense AI, an AI-powered semantic screenshot search project using natural-language queries.
- Explored OCR, text processing, semantic search, and multilingual AI concepts to make screenshot content searchable.
- Applied problem-solving and debugging while developing and testing the project workflow.

EDUCATION

Saranathan College of Engineering
B.Tech, Information Technology (2023 – 2027) – CGPA: 8.3/10

Chidhambaram Matriculation Higher Secondary School
Higher Secondary (HSC) (2022 – 2023) – 91%

CERTIFICATIONS

- Cisco Networking Basics
- GeeksforGeeks – IP Addressing

EXPERIENCE

Fresher

SOFT SKILLS

Problem Solving • Analytical Thinking • Logical Reasoning • Debugging • Team Collaboration • Communication • Quick Learning • Adaptability

LANGUAGES

English (Fluent), Tamil (Native)